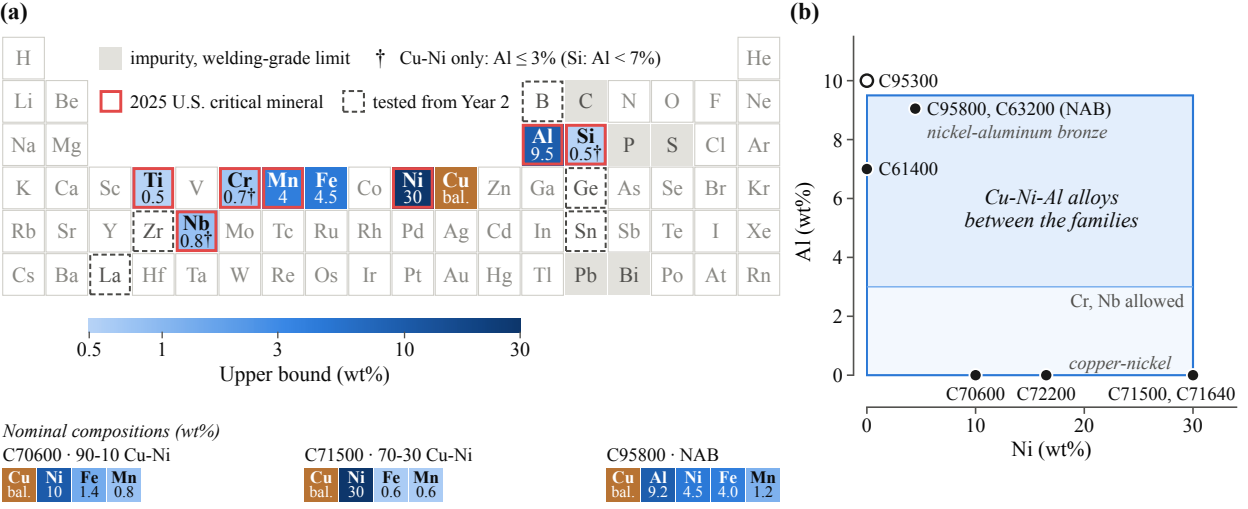

(a)
impurity, welding-grade limit
† Cu-Ni only: Al ≤ 3% (Si: Al < 7%)
2025 U.S. critical mineral
tested from Year 2
Ti 0.5
Cr 0.7†
Mn 4
Fe 4.5
Ni 30
Cu bal.
Al 9.5
Si 0.5†
Nb 0.8†
0.5
1
3
10
30
Upper bound (wt%)
Nominal compositions (wt%)
C70600 · 90-10 Cu-Ni
Cu bal. Ni 10 Fe 1.4 Mn 0.8
C71500 · 70-30 Cu-Ni
Cu bal. Ni 30 Fe 0.6 Mn 0.6
C95800 · NAB
Cu bal. Al 9.2 Ni 4.5 Fe 4.0 Mn 1.2
(b)
C95300
C95800, C63200 (NAB)
nickel-aluminum bronze
C61400
Cu-Ni-Al alloys
between the families
Cr, Nb allowed
copper-nickel
C70600
C72200
C71500, C71640
Al (wt%)
Ni (wt%)
0
2
4
6
8
10
0
10
20
30